9-Equilibrium Calculations

You have seen that the magnitude of an equilibrium constant gives us some feel for the extent to which the reaction proceeds at equilibrium. Sometimes however, it is necessary to have more than merely a qualitative knowledge equilibium concentrations. This requires that we be able to use the equilibrium law for purposes of calculation.

Equilibrium calculations for gaseous reactions can be performed using either K_p or K_c but for reactions in solution we must use K_c.

Whether we with concentrations or partial pressures, however, the same basic principles apply.

Overall, we can divide equilibrium calculations into two main categories

1. Calculating equilibrium constants from known equilibrium concentrations or partial pressures.
2. Calculating one or more equilibrium concentrations or partial pressures use the known value of K_c or K_p.

1-Calculating K_c from Equilibrium Concentrations

You've seen that we can obtain equilibrium constants from thermodynamic data. However, another way to determine the value of K_c is to carry out reaction, measure the concentrations of reactants and products after equilibrium has been reached, and then use the equilibrium values in the equilibrium law to compute K_c. As an example, let's look again at the decomposition of N_2O_4.

We saw that

if 0.0350 mol of N_2O_4 is placed into a 1 L flask at 25 °C, the concentrations of N_2O_4 and NO_2 at equilibrium are

$$[N_2O_4] = 0.0292 \text{ mol/L}$$

$$[NO_2] = 0.0116 \text{ mol/L}$$

To calculate K_c for this reaction, we substitute the equilibrium concentrations into

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the mass action expression of the equilibrium law.

$$\frac{[NO_2]^2}{[N_2O_4]} = K_c$$

$$\frac{(0.0116)^2}{(0.0292)} = K_c$$

Performing the arithmetic gives:

$$K_c = 4.61 \times 10^{-3}$$

Although calculating an equilibrium constant this way is easy, sometimes we have to do a little work to figure out what all the concentrations are, as shown in Example.11.

EX.11

At a certain temperature, a mixture of H_2 and I_2 was prepared by placing 0.2 mol of H_2 and 0.2 mol of I_2 into a 2-liter flask.

After a period of time the equilibrium: $H_2(g) + I_2(g) = 2HI(g)$

was established. At equilibrium, the I_2 concentration had dropped to 0.02 mol/L. What is the value of K_c for this reaction at this temperature?

Analysis

The first step in any equilibrium problem is to write the balance chemical equation and the related equilibrium law. The equation is already given, and the equilibrium law corresponding to it is:

$$\frac{[HI]^2}{[H_2][I_2]} = K_c$$

To calculate the value of K_c **we must substitute the equilibrium concentrations of H_2, I_2 and HI into the mass action expression.**

The system starts with a set of **initial concentrations**, and to reach equilibrium a chemical change occurs. There is no HI present initially, so the reaction above must proceed to the right, because there has to be some HI present for the system

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be at equilibrium.This change will increase the HI concentration and decrease the concentrations of H_2 and I_2. If we can determine what the changes are, we can figure out the equilibrium concentrations. In fact, **the key to solving** almost all equilibrium problems is **determining how the concentrations change as the system comes to equilibrium.**

To help us in our analysis **we will construct a concentration table** beneath the chemical equation. We will do this for most of the equilibrium problems we encounter from now on, so let's examine the general form of the table.

In the first row we write the initial molar concentrations of the reactants and products. (All entries in the table will be molar concentrations, not moles. These are the only quantities that satisfy the equilibrium law.) Then, **in the second row we enter the changes in concentration, using a positive sign if the concentration is creasing and a negative sign if it is decreasing.** Finally, **adding the changes to the initial concentrations gives the equilibrium concent -rations.**

(equilibrium concentration)	**=**	**(initial concentration)**	**+ (change in concentration)**

To set up the concentration table, it is essential that you realize that the changes in concentration must be in the same ratio as the coefficients of the balanced equation. This is because the changes are caused by the chemical reaction proceeding in one direction or the other. In this problem, the coefficients of H_2 and I_2 are the same, so as the reaction proceeds to the right, the concentrations of H_2 and I_2 must decrease by the same amount. Similarly, because the coefficient of HI is twice that of H_2 or I_2 the concentration of HI must increase by twice as much.

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Solution

To construct the table, we begin by entering the data given in the statement of the problem. This is shown in bold type. The values in regular type are:

Initial Concentrations: Notice that we have calculated the ratio of moles to liter's for both the H_2 and I_2 ,(0.200 mol/2.00 L) = 0.100 M. Because no HI was placed the reaction mixture, its initial concentration has been set to zero. **Changes in Concentration**: We have been given both the initial and equilibrium concentrations of I_2, so by difference we can calculate the change for I_2.

Equilibrium Concentrations: For H_2 and HI, we just add the change to the initial value.

Now we can substitute the equilibrium concentrations into the mass action expression and calculate K_c.

$$K_c = \frac{(0.160)^2}{(0.020)\,(0.020)}$$
$$= 64$$

The Concentration Table

Let's review some key points that apply not only to this problem, but to others that deal with equilibrium calculations.

1. The only values that we can substitute into the mass action expression in equilibrium law are equilibrium concentrations— the values that appear the last row of the table.

2. When we enter initial concentrations into the table, they should be in units of moles per liter (mol/L). The initial concentrations are those present in the reaction mixture when it's prepared; we imagine that no reaction occurs until everything is mixed.

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3. The changes in concentration always occur in the same ratio as the coeffici - ents in the balanced equation.

For example, if we were dealing with the equilibrium $3H_2(g) + N_2(g) = 2NH_3(g)$ and found that the $N_2(g)$ concentration decreases by 0.10M during the approach to equilibrium, the entries in the "change" row would be as follows:

	$3H_2(g)$	$+ N_2(g)$	$= 2NH_3(g)$
Change in concentration:	**-3 x (0.1M)**	**-1 x (0.1M)**	**-2 x (0.1M)**
	- 0.3M	**- 0.1M**	**- 0.2M**

Practice Exercise 7

In a particular experiment, it was found that when $O_2(g)$ and $CO(g)$were mixed and reacted according to the equation

$$2CO(g) + O_2(g) = 2CO_2(g)$$

the O_2 concentration decreased by 0.03 mol/L when the reaction reached equilibr- ium. How had the concentrations of CO and CO_2 changed?

Practice Exercise 8

An equilibrium was established for the reaction:

$$CO(g) + H_2O(g) = CO_2(g) + H_2(g)$$

at 500 C (This is an industrially important reaction for the preparation of hydrogen) At equilibrium, the following concentrations were found in the reaction vessel: $[CO] = 0.18M$, $[H_2O] = 0.0411M$, $[CO_2] = 0.15M$ and $[H_2] = 0.2M$. What is the value of K_c for this reaction?

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2-Calculating Equilibrium Concentrations Using K_c

In the simplest calculation of this type, all but one of the equilibrium concentrations are known, as illustrated in Example.12.

EX.12

The reversible reaction

$$CH_4(g) + H_2O(g) = CO(g) + 3H_2(g)$$

has been used as a commercial source of hydrogen. At 1500°C, an equilibrium mix -ture of these gases was found to have the following concentrations:

$[CO] = 0.3M$, $[H_2] = 0.8$ M, and $[CH_4] = 0.4M$. At 1500°C, $K = 5.67$ for this reaction. What was the equilibrium concentration of H_2O (g) in this mixture?

Solution

The first step, once we have the chemical equation for the equilibrium, write the equilibrium law for the reaction.

$$K_c = \frac{[CO]\,[H_2]^3}{[CH_4]\,[H_2O]}$$

The equilibrium constant and all of the equilibrium concentrations except that for H_2O are known, so we substitute these values into the equilibrium law .

$$5.67 = \frac{(0.300)\,(0.800)^3}{(0.400)\,[H_2O]}$$

Solving for $[H_2O]$ gives

$$[H_2O] = \frac{(0.300)\,(0.800)^3}{(0.400)\,(5.67)}$$

$$= 0.0677\ M$$

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3-Calculating Equilibrium Concentrations Using K_c and Initial Concentrations

A more complex type of calculation involves the use of initial concentration and K_c to compute equilibrium concentrations. Although some of these problems can be so complicated that a computer is needed to solve them we learn the general principles involved by working on simple calculations.

EX.13

The reaction

$$CO(g) + H_2O(g) = CO_2(g) + H_2(g)$$

has $K_c = 4.06$ at 500 °C. If 0.1 mol of CO and 0.1 mol of H_2O (g) are placed in a 1L reaction vessel at this temperature, what are the concentrations of the react -ants and products when the system reaches equilibrium?

Analysis The key to solving this kind of problem is recognizing that at equilib - rium the mass action expression must equal K_c.

$$\frac{[CO_2][H_2]}{[CO][H_2O]} = 4.06 \qquad \text{(at equilibrium)}$$

We must find values for the concentrations that satisfy this condition. Because we don't know what these concentrations are, we represent them a basically as unknowns . This is where the concentration table is very helpful. build the table, we need quantities to enter into the **"initial concentration" "changes in concentration,"** and **"equilibrium concentrations"** rows.

Initial Concentrations The initial concentrations of CO and H_2O are each 0.1 mol/1L = 0.1 M. Since no CO_2 or H_2 are initially placed into the reaction vessel, their initial concentrations both are zero.

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Changes in Concentration Some CO_2 and H_2 must form for the reaction to reach equilibrium. This also means that some CO and H_2O must react. But how much? If we knew the answer, we could calculate the equilibrium concentrations. Therefore,

the changes in concentration are our unknown quantities.

Let us allow x to be equal to the number of moles per liter of CO that react.The change in the concentration of CO is then -x (it is negative because the change decr -eases the CO concentration). Because CO and H_2 react in a 1: 1 mole ratio, the change in the H_2O concentration is also -x. Since 1 mol each of CO_2 and H_2 is formed from 1 mol of CO, the CO_2 and H_2 concentrations each increase by x (their changes are + x).

Equilibrium Concentrations We obtain the equilibrium concentrations as

(equilibrium concentration) = (initial concentration) + (change in concentration)

Solution Here is the completed concentration table.

	$CO(g)$	$+ H_2O(g)$	$= CO_2(g)$	$+ H_2(g)$
Initial concentration(M)	**0.1**	**0.1**	**0.0**	**0.0**
Change in concentration(M)	**-x**	**-x**	**+x**	**+x**
Equilibrium concentration(M)	**0.1-x**	**0.1-x**	**x**	**x**

Note that the last line in the table tells us that the equilibrium CO and H_2O concen -trations are equal to the number of moles per liter that were present initially minus the number of moles per liter that react. The equilibrium concentrations of CO_2 and H_2 equal the number of moles per liter of each that forms, since no CO_2 or H_2 are present initially.

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Next, we substitute the quantities from the "equilibrium concentration" row into the mass action expression and solve for x.

$$\frac{(x)\,(x)}{(0.100 - x)\,(0.100 - x)} = 4.06$$

which we can write as:

$$\frac{x^2}{(0.100 - x)^2} = 4.06$$

In this problem we can solve the equation for x most easily by taking the square root of both sides.

$$\frac{x}{(0.100 - x)} = \sqrt{4.06} = 2.01$$

Clearing fractions gives $x = 2.01\,(0.1 - x)$ $x = 0.201 - 2.01x$

Collecting terms in x gives: $x + 2.01x = 0.201$

$3.01x = 0.201$ $x = 0.0668$

Now that we know the value of x, we can calculate the equilibrium concentration from the last row of the table.

$[CO] = 0.1 - x = 0.1 - 0.0668\ 0.033$ M

$[H_2O] = 0.1 - x = 0.1 - 0.0668 = 0.033$ M

$[CO_2] = x = 0.0668$ M

$[H_2] = x = 0.0668$ M

Check the answer: One way to check the answers is to substitute the equilibrium concentrations we've found into the mass action expression and evaluate reaction quotient. If our answers are correct, Q should equal K_c. Let's do this.

$$Q = \frac{(0.0668)^2}{(0.033)^2} = 4.1$$

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Rounding K_c to two significant figures gives 4.1, so the calculated concentrations satisfy the equilibrium law.

EX.14

In the preceding example it was stated that the reaction

$$CO(g) + H_2O(g) = CO_2(g) + H_2(g)$$

has $K_c = 4.06$ at 500 °C. Suppose 0.06 mol each of CO and H_2O are mixed with 0.1 mol each of CO_2 and H_2 in a 1L reaction vessel. hat will the concentrations of all the substances be when the mixture reaches equilibrium at this temperature?

Analysis

We will proceed in much the same way as in the preceding example. However, this time determining the algebraic signs of x will not be quite as simple because none of the initial concentrations is zero. The best way to determine algebraic signs is to use the initial concentrations to calculate the initial reaction quotient. Then we can compare Q to K and, by reasoning, figure out which was the reaction must proceed to make Q equal to K_c.

Solution

The equilibrium law for the reaction is

$$\frac{[CO_2][H_2]}{[CO][H_2O]} = K_c$$

Let's use the initial concentrations, shown in the first row of the concentration table below, to determine the initial value of the reaction quotient.

$$Q_{initial} = \frac{(0.100)(0.100)}{(0.0600)(0.0600)} = 2.78 < K_c$$

As indicated, $Q_{initial}$ is less than K_c, so the system is not at equilibrium. To reach equilibrium Q must become larger, which requires an increase in the concentrations of CO_2 and H_2 as the reaction proceeds. This means that for CO_2 and H_2 must be positive, and therefore, for CO and H_2, x must be negative.

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Here is the completed concentration table.

	$CO(g)$ +	$H_2O(g)$ =	$CO_2(g)$ +	$H_2(g)$
Initial concentration(M)	**0.06**	**0.06**	**0.1**	**0.1**
Change in concentration(M)	**-x**	**-x**	**+x**	**+x**
Equilibrium concentration(M)	**0.06-x**	**0.06-x**	**0.1+ x**	**0.1+ x**

Substituting equilibrium quantities into the mass action expression in the equilibrium law gives us

$$\frac{(0.100 + x)^2}{(0.0600 - x)^2} = 4.06$$

Taking the square root of both sides yields

$$\frac{0.100 + x}{0.0600 - x} = 2.01$$

To solve for x we first multiply each side by 0.0600 - x to obtain

$$0.1 + x = 2.01\ (0.06 - x)$$

$$0.1 + x = 0.121 - 2.01x$$

Collecting terms in x to one side and the constants to the other gives

$$x + 2.01x = 0.121 - 0.1$$

$$3.01x = 0.021$$

$$x = 0.007$$

Now we can calculate the equilibrium concentrations:

$[CO]\ [H_2O] = (0.06 - x) = 0.06 - 0.007 = 0.053M$

$[CO_2] = [H_2] = (0.1 + x) = 0.1 + 0.007 = 0.107M$

Check Answer

As a check, let's evaluate the reaction quotient using the calculated equilibrium concentrations.

$$Q = \frac{(0.107)^2}{(0.0530)^2} = 4.08$$

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This is acceptably close to the value of K_c. (That it is not exactly equal to K_c is because of the rounding of answers during the calculations.)

EX.15

At a certain temperature $K_c = 4.5$ for the reaction

$$N_2O_4(g) = 2NO_2(g)$$

If 0.3 mol of N_2O_4 is placed into a 2L container at this temperature, what will the equilibrium concentrations of both gases?

Analysis As in the preceding example, at equilibrium the mass action express- ion must be equal to K_c.

$$\frac{[NO_2]^2}{[N_2O_4]} = 4.50$$

We will need to find algebraic expressions for the equilibrium concentrations and substitute them into the mass action expression. To obtain these, we set up the concentration table for the reaction.

Initial Concentrations The initial concentration of N_2O_4 is 0.3 mol/2 L= 0.15M Since no NO_2 was placed in the reaction vessel, its initial concentration is **0.0 M**

Changes in Concentrations There is no NO_2 in the reaction mixture, so we know concentration must increase. This means the N_2O_4 concentration must decrea some of the NO_2 is formed. Let's allow x to be the number of moles per lite N_2O_4 that reacts, so the change in the N_2O_4 concentration is -x. Because of stoichiometry of the reaction, the NO_2 concentration must increase by 2x, so change in concentra- tion is +2x.

Equilibrium Concentrations As before, we add the change to the initial concentration in each column to obtain expressions for the equilibrium concentrations

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Solution: Here is the concentration table.

	$N_2O_4(g)$ =	$2NO_2(g)$
Initial concentration(M)	**0.15**	**0.0**
Change in concentration(M)	**-x**	**+2x**
Equilibrium concentration(M)	**0.15-x**	**2x**

Now we substitute the equilibrium quantities into the mass action expression.

$$\frac{(2x)^2}{(0.150 - x)} = 4.50$$

or

$$\frac{4x^2}{(0.150 - x)} = 4.50$$

This time the left side of the equation is not a perfect square, so we cannot just take the square root of both sides as in Example.13. However, because equation involves terms in x^2, x, and a constant, we can use the quadratic formula. Recall that for a quadratic equation of the form

$$ax^2 + bx + c = 0$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Expanding our equation above gives:

$$4x^2 = 4.50(0.150 - x)$$
$$= 0.675 - 4.50x$$

Arranging terms in the standard order gives

$$\mathbf{4x^2 + 4.50x - 0.675 = 0}$$

Therefore, the quantities we will substitute into the quadratic formula are as lows: a = 4, b = 4.50, and c = - 0.675. Making these substitutions gives:

$$x = \frac{-4.50 \pm \sqrt{(4.50)^2 - 4(4)(-0.675)}}{2(4)}$$

$$= \frac{-4.50 \pm \sqrt{31.05}}{8}$$

$$= \frac{-4.50 \pm 5.57}{8}$$

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Because of the ± term, there are two values of x that satisfy the equation, $x = 0.134$, and $x = -1.26$. However, only the first value, $x = 0.134$, makes any sense chemically. Using this value, the equilibrium concentrations are:

$$[N_2O_4] = 0.15 - 0.134 = 0.016 \text{ M}$$

$$[NO_2] = 2(0.134) = 0.268 \text{ M}$$

Notice that if we had used the negative root, -1.26, both equilibrium concentrations would be negative. Negative concentrations are impossible, so $x = -1.26$ is not acceptable for chemical reasons. In general, whenever you use the quadratic equation in a chemical calculation one root will be satisfactory and the other will lead to answers that are nonsense.

Check the answer: Once again, we can evaluate the reaction quotient using the calculated equilibrium values. When we do this, we obtain $Q = 4.49$, which is acceptably close to the value of K_c given.

Equilibrium problems can be even more complex than the one we have just discussed; however, you will not have to deal with them. In fact, in most of the problems you encounter there will be ways of simplifing the algebra. Only rarely will it even be necessary to resort to the quadratic equation.

Practice Exercise 9

During an experiment, 0.2 mol of H_2 and 0.2 mol of I_2 were placed into a 1 liter vessel where the reaction:

$$H_2(g) + I_2(g) = 2HI(g)$$

came to equilibrium. For this reaction, $K_c = 49.5$ at the temperature of the experiment. What were the equilibrium concentrations of H_2, I_2 and HI?

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